English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021 Test 1 *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (1 point) At a shooting gallery, a boy pays 2 BGN for three shots. The prizes are 10 BGN for three hits, 5 for two hits, and 1 for one hit. His hit probability per shot is $1/3$. Is the game fair?[1] For which values of the largest prize is the game favourable to him? If the boy makes a profit, what is the probability he hit the target on his third shot?

Problem 2. The first box contains 4 white and 6 black balls; the second 7 white and 3 black. Draw one ball randomly from each, add one white ball, and place all three in a third box.

1. (0.4 points) What is the probability that a random ball from the third box is white?
2. (0.6 points) Draw from the third box with replacement until the first white ball. Let X count draws. Find $\mathbb{E}X$. Is X geometric?

Problem 3. (1 point) A bag has $n \in \mathbb{N}$ coins with values $1, \dots, n$. A person takes a handful, with each coin selected with probability $p \in (0,1)$. Find the expected sum S.

Problem 4.
1. (0.4 points) State when events $A, B \subset \Omega$ are independent and when they are disjoint. Can they be independent if (a) disjoint, or (b) $A \subset B$? If so, under what conditions?
2. (0.6 points) State when discrete random variables X, Y are independent and when identically distributed. Let X, Y, Z be identically distributed and pairwise independent. Assume ties have probability zero. Find $\mathbb{P}(X < Y)$, $\mathbb{P}(Y < Z)$ and $\mathbb{P}(Z < X)$. State inclusion–exclusion for three events. Prove $a = \min\{\mathbb{P}(X < Y), \mathbb{P}(Y < Z), \mathbb{P}(Z < X)\} \le 2/3$.[2]

[1] A game is fair if expected net winnings are zero.

[2] Can you construct equality? A simple example uses suitable probabilities on events of the form $(X = n_1, Y = n_2, Z = n_3)$.